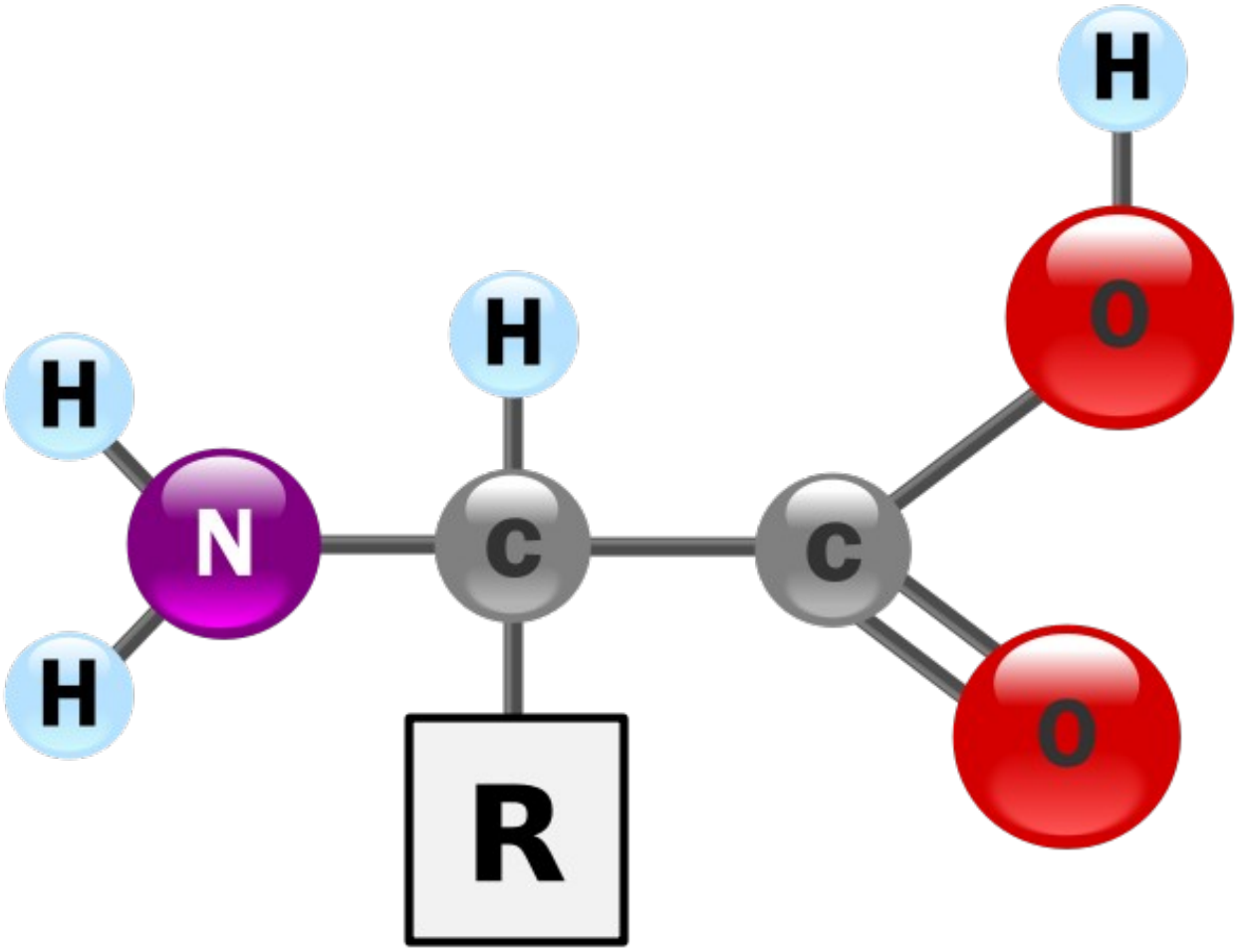
AMINO ACIDS



Learning Objectives

At the end of this session, student will be able to learn and understand;

1.What are amino acids and their role in protein synthesis?

2.Different types of amino acids and their natural roles.

3.Nutritional classification of amino acids

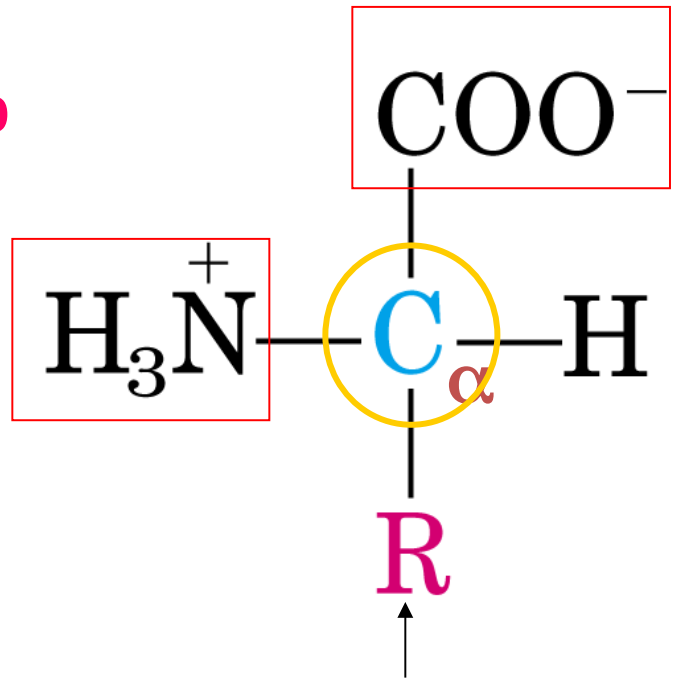
Amino acids

- Amino acids are the building blocks of protein.
- Protein of all living species are made up of same set of standard 20 amino acids.
  - Standard α-amino acids:
    - Only *20 are commonly found as constituents of proteins*
  - Nonstandard amino acids:
    - Amino acid derivatives found in protein
    - Non-protein amino acid

General structure of α-amino acids

- carboxyl group

amino group

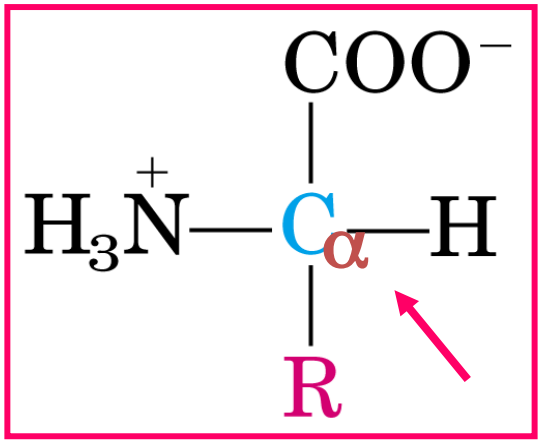


side chain

(or R group, R="Remainder of the molecule")

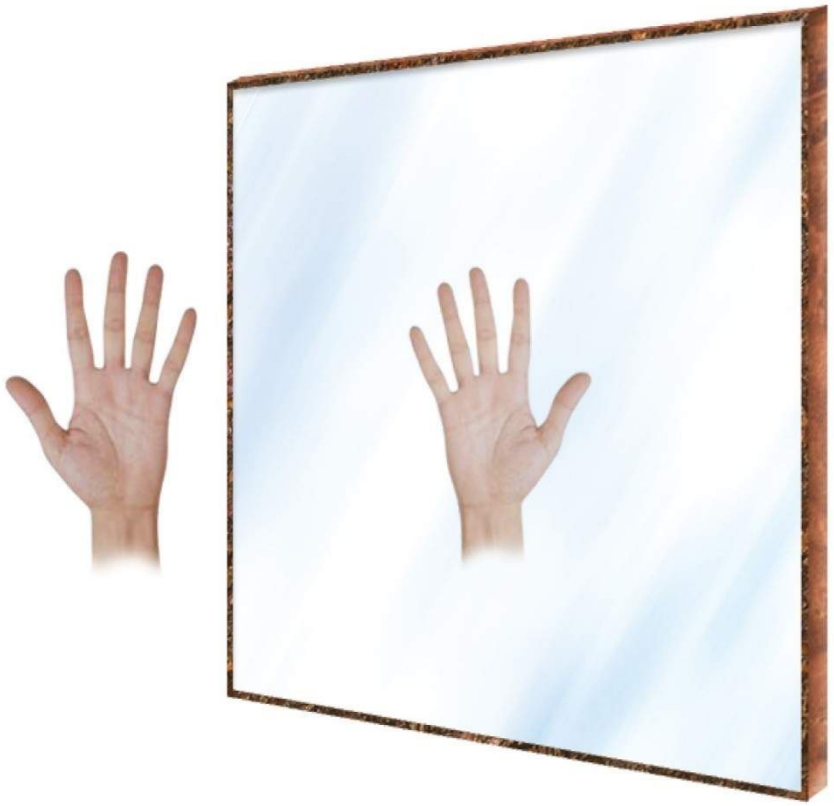
Stereochemistry of amino acids

- Configuration
- For all the common amino acids except Glycine, the α-carbon is bonded to four different groups.
- The α-carbon atom is thus a chiral center.



Chiral

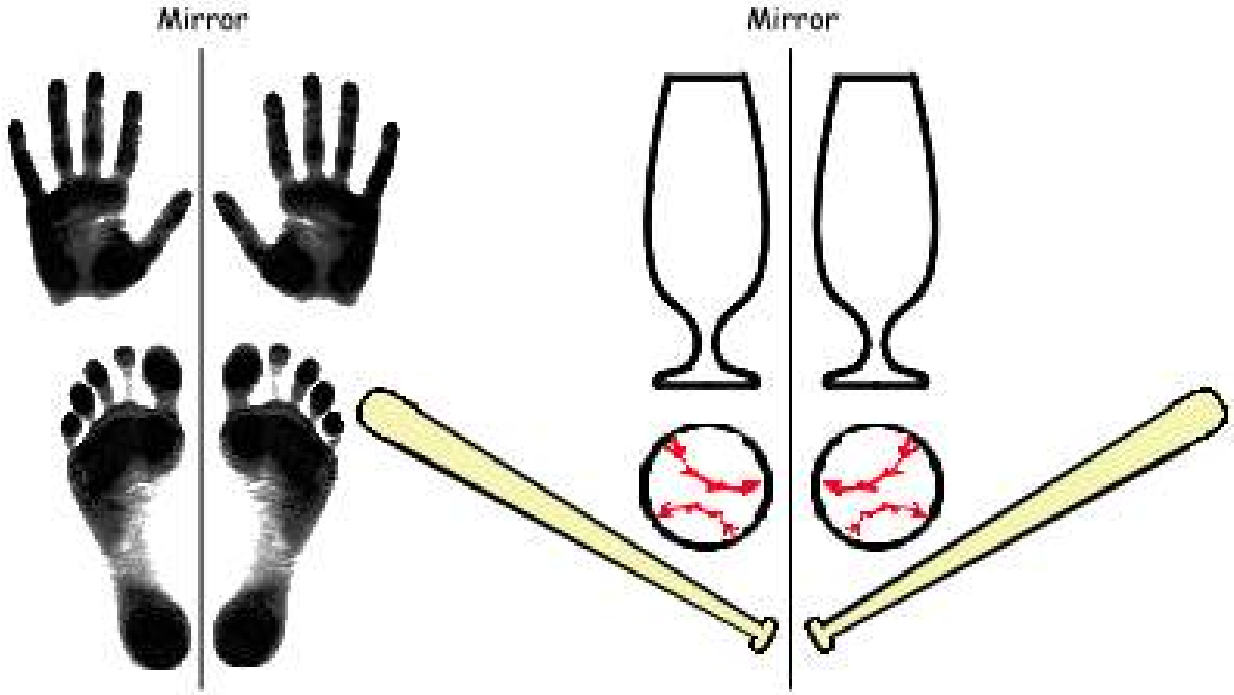
- Chiral (from Greek *cheir*, meaning “hand”):
  - An object or a system cannot be superimposed on its mirror image.
  - One hand does not match the other when superimposed.



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CHIRALITY

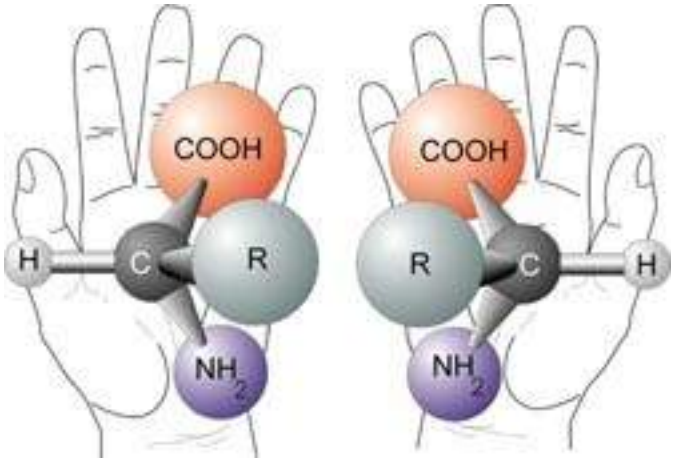
An object that cannot be superimposed on its mirror image is called chiral



Chiral objects
Nonsuperimposable
mirror images

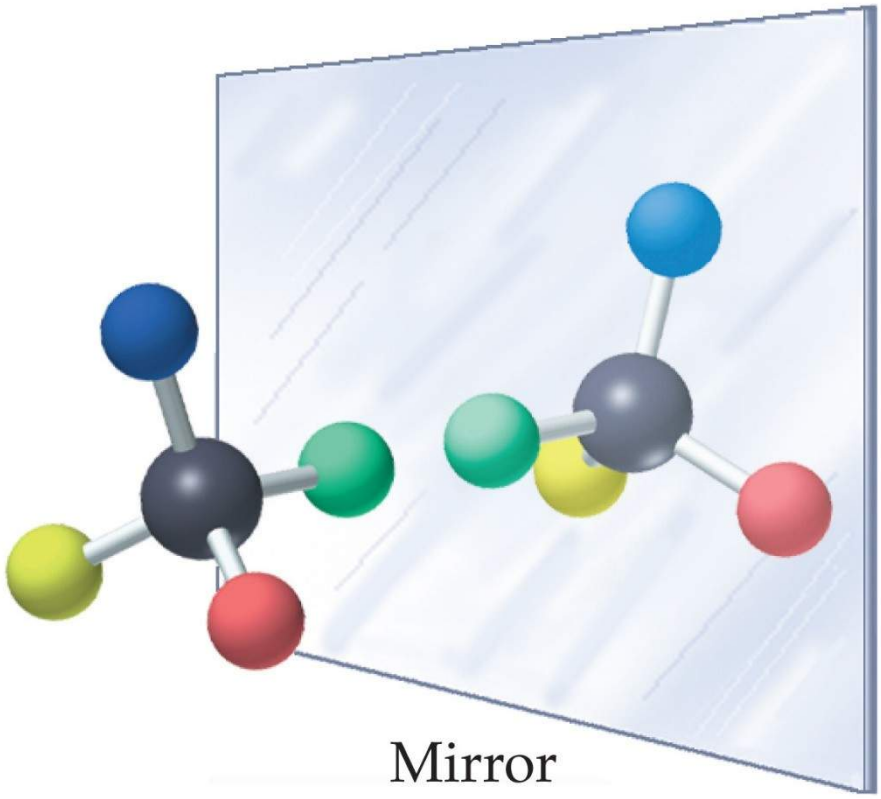
Nonchiral objects
Superimposable
mirror images

Chiral molecule



- Chiral molecule:
  - if a molecule has an atom bonded to four different groups, it can be chiral.

- Enantiomers:
  - a chiral molecule, has left- and right-handed isomers, called enantiomers.

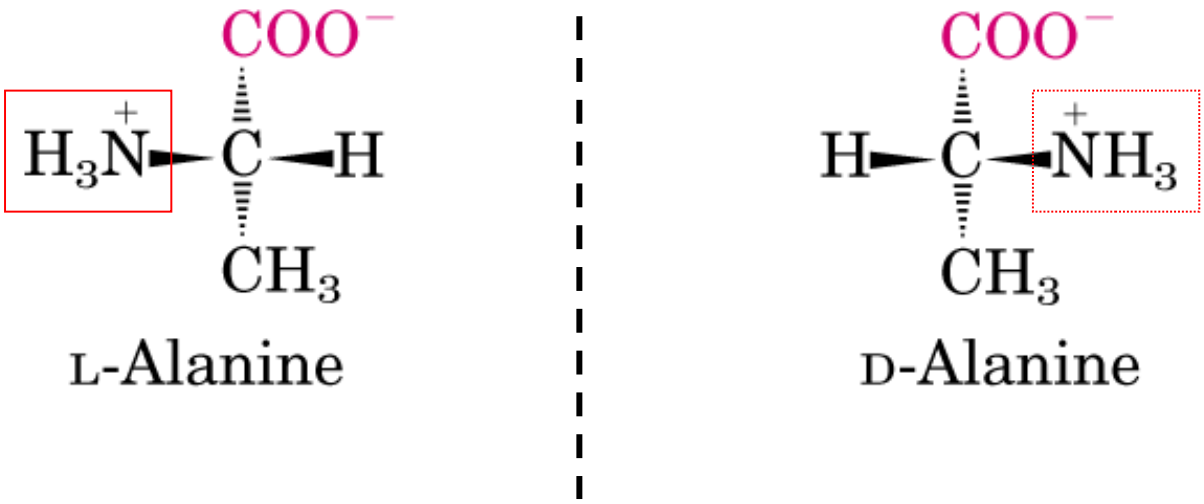


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- The two enantiomers of each amino acid are designated by D,L system according to the

  D- and L-glyceraldehyde.

- D: Dextrorotation; L: Levorotation



- **Only the *L-amino acids* have been found in proteins.**
- (D-isomers have been found only in small peptides of bacteria cell walls or in some peptide antibiotics).

The Classification of 20 standard Amino Acids

- The name and abbreviation of amino acids
  - All the AAs were given a trivial (common) name.
    - Glutamate from wheat gluten.
    - Tyrosine from cheese (“tyros” in Greek).
  - Each AA is given a 3 letter abbreviation and 1 letter symbol.
    - They often the first three letter and the first letter. When there is confusion, an alternative is used.
- They should be remembered !

TABLE 5.1

Names and Abbreviations of the Standard Amino Acids

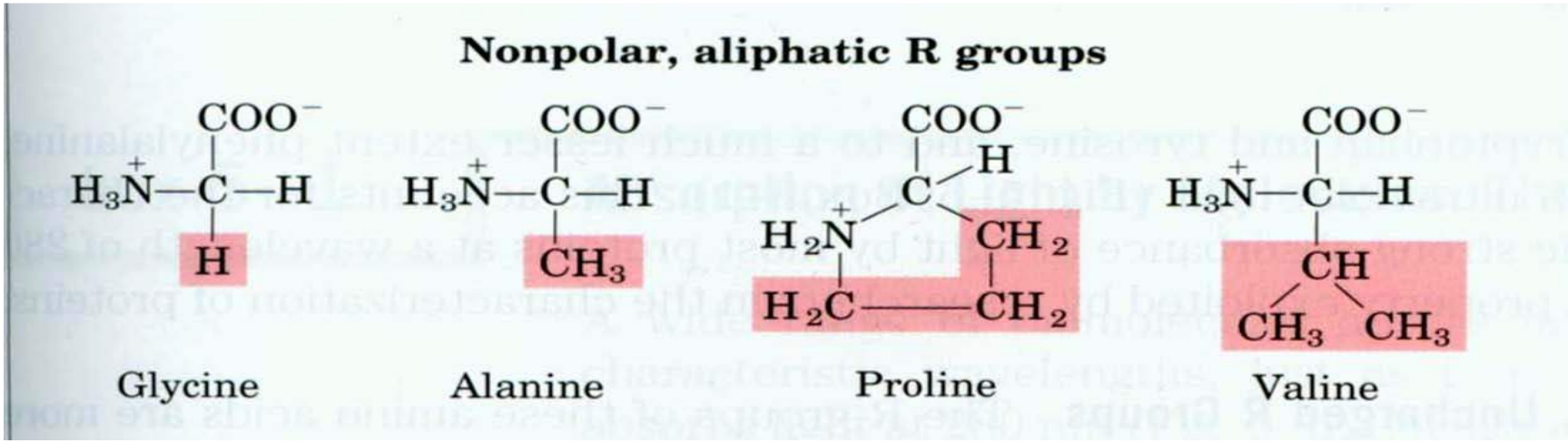
Amino Acid	Three-Letter Abbreviations	One-Letter Abbreviations
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

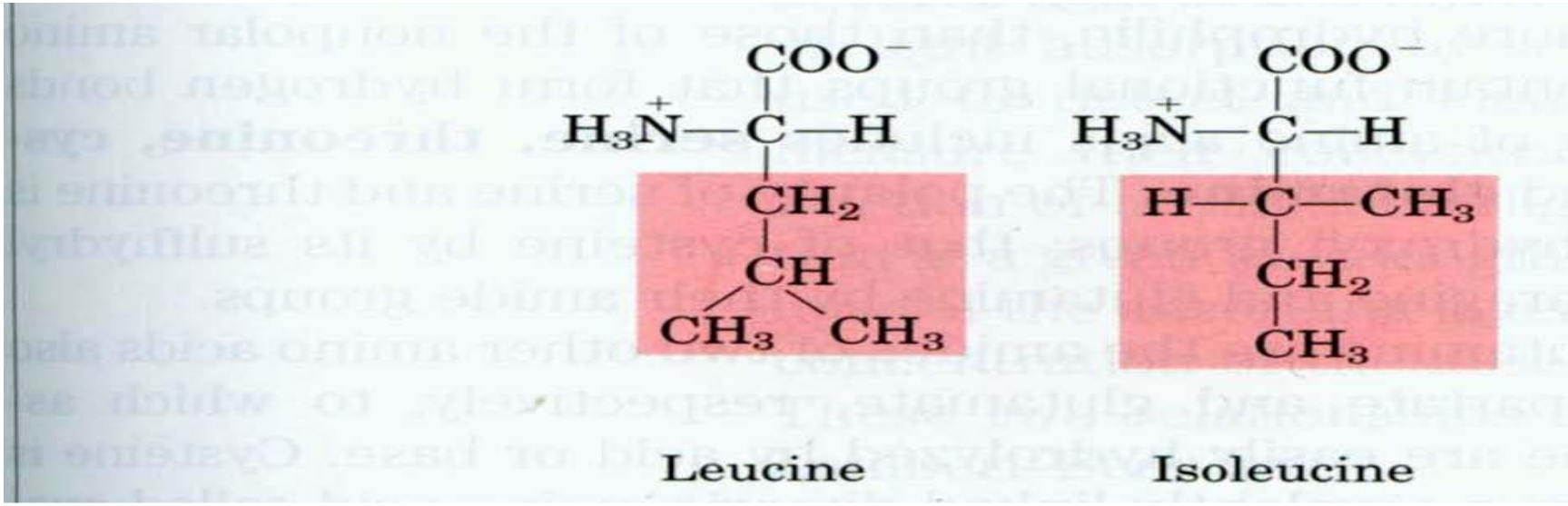
Classification of Amino Acids

- Based on polarity
  - ***Nonpolar, aliphatic amino acids***
  - ***Polar, uncharged amino acids***
  - ***Aromatic amino acids***
  - ***Acidic amino acids***
  - ***Basic amino acids***

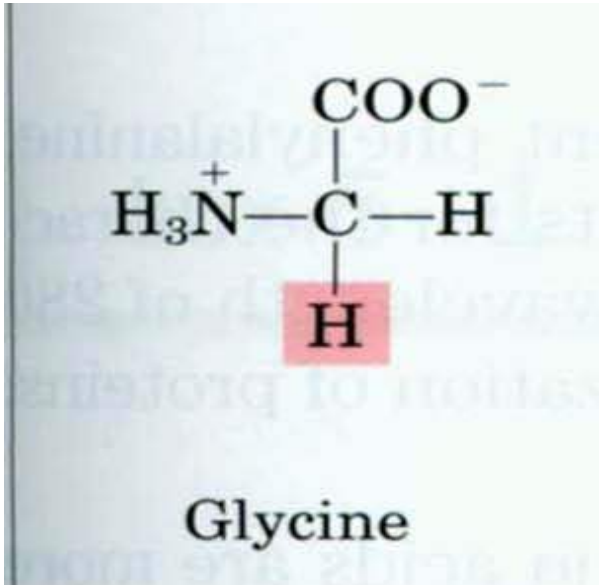
NONPOLAR, ALIPHATIC AMINO ACIDS

- **Gly, Ala, Pro, Val, Leu, and Ile**



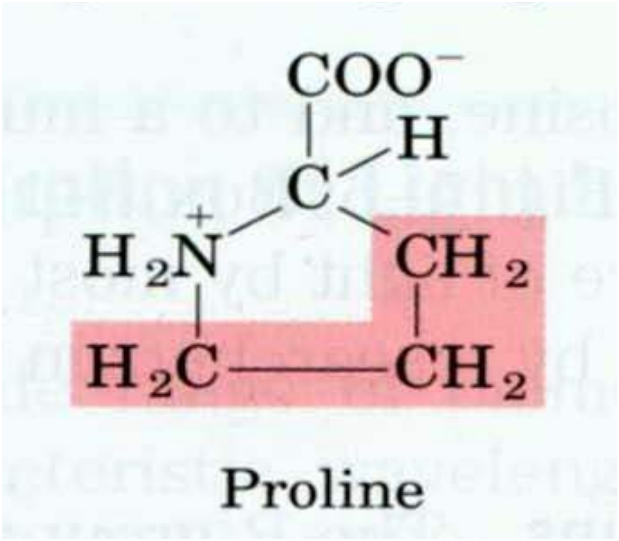


- Gly:
  - R group: hydrogen.
  - symmetric, not chiral.



Glycine

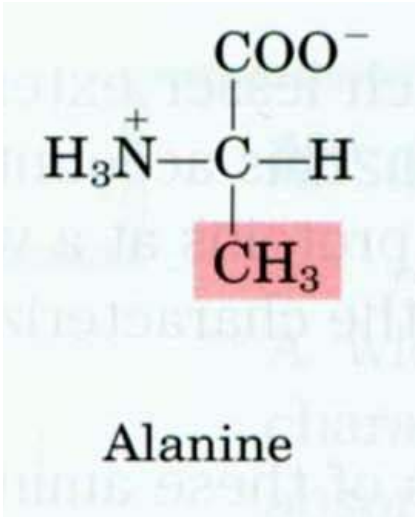
- Pro:
  - *Imino acid*.
  - five-membered ring structure, rigid in conformation.



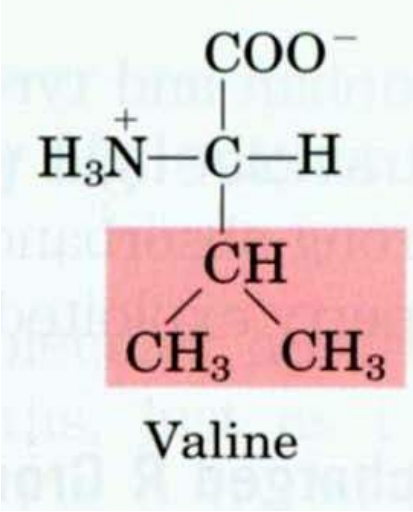
Proline

- **Ala, Val, Leu, and Ile**

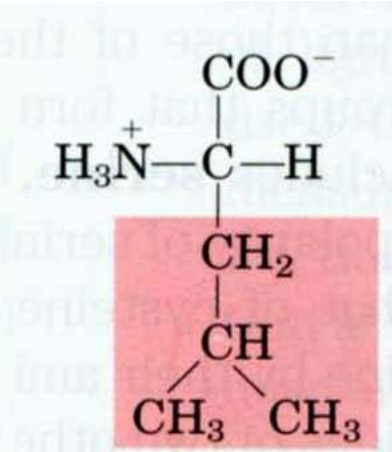
- Hydrocarbon R groups, often involved in hydrophobic interactions for stabilizing protein structure .



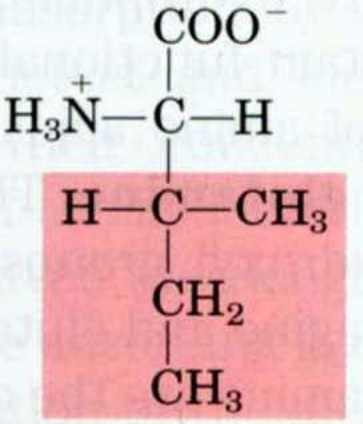
Alanine



Valine



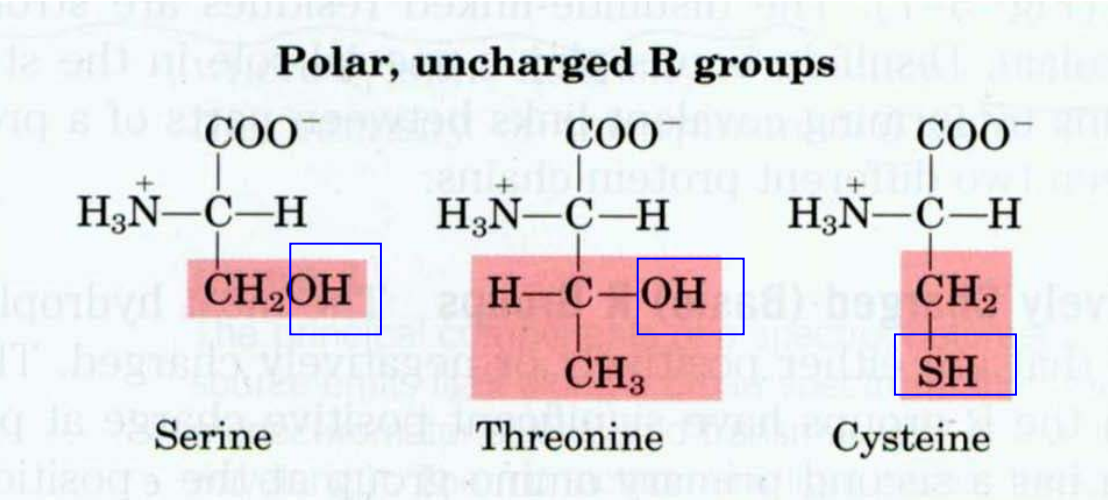
Leucine

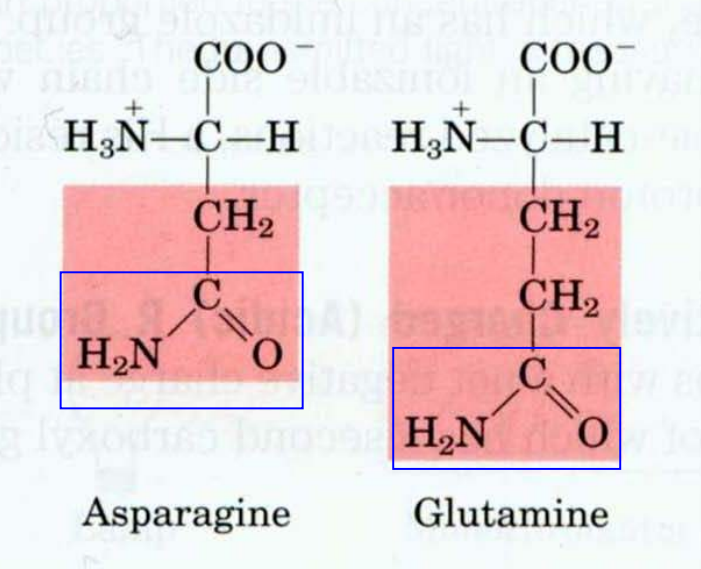


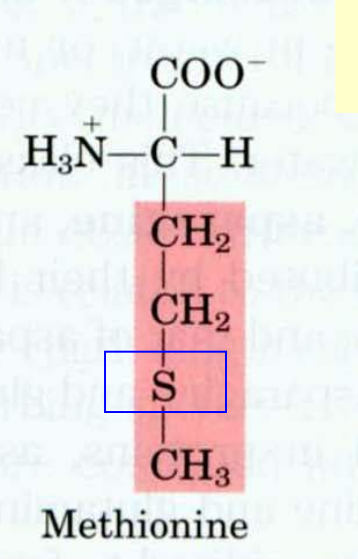
Isoleucine

Polar, uncharged amino acids

- **Ser, Thr, Cys, Asn, Gln and Met.**



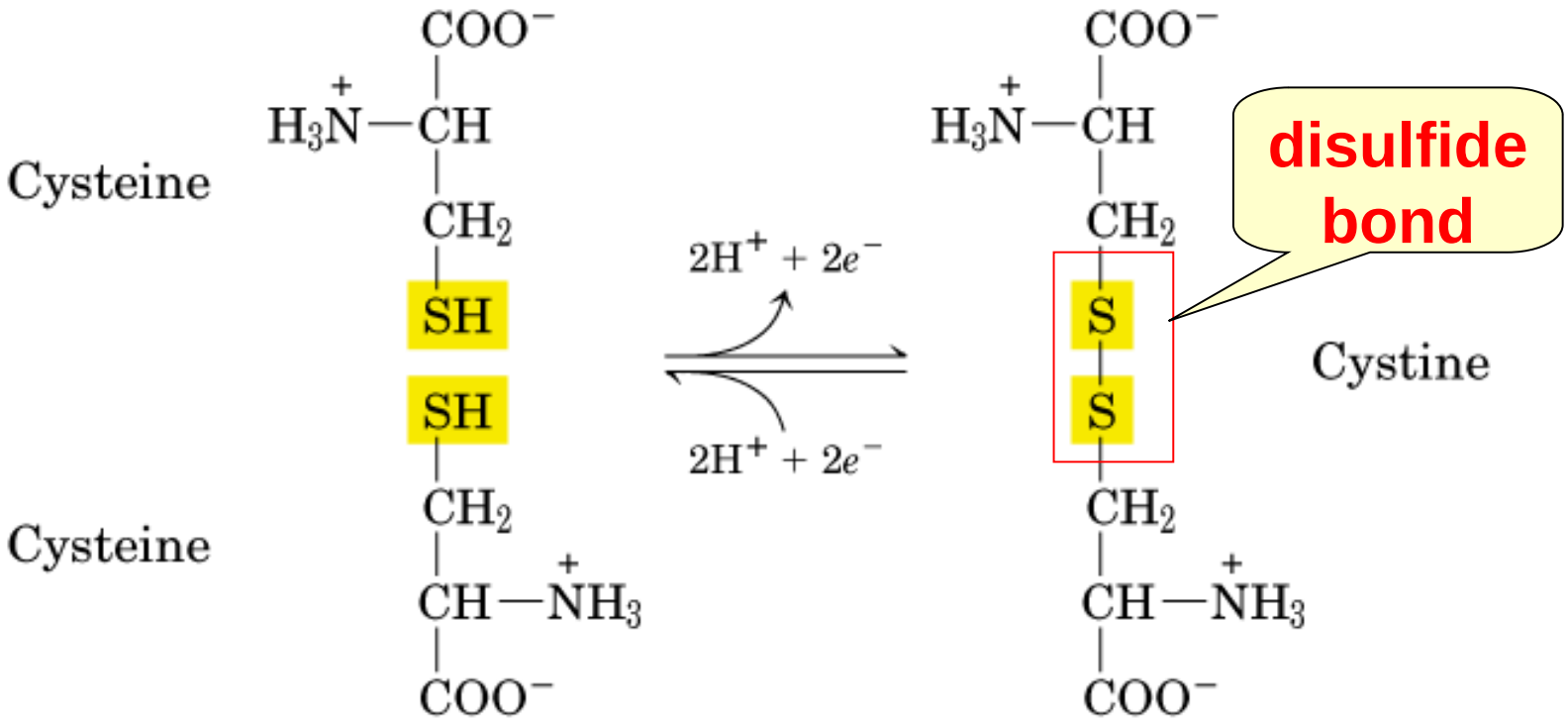




- **R group:**
Hydrophilic
  - Hydroxyl groups
  - Sulfur atoms
  - Amide groups

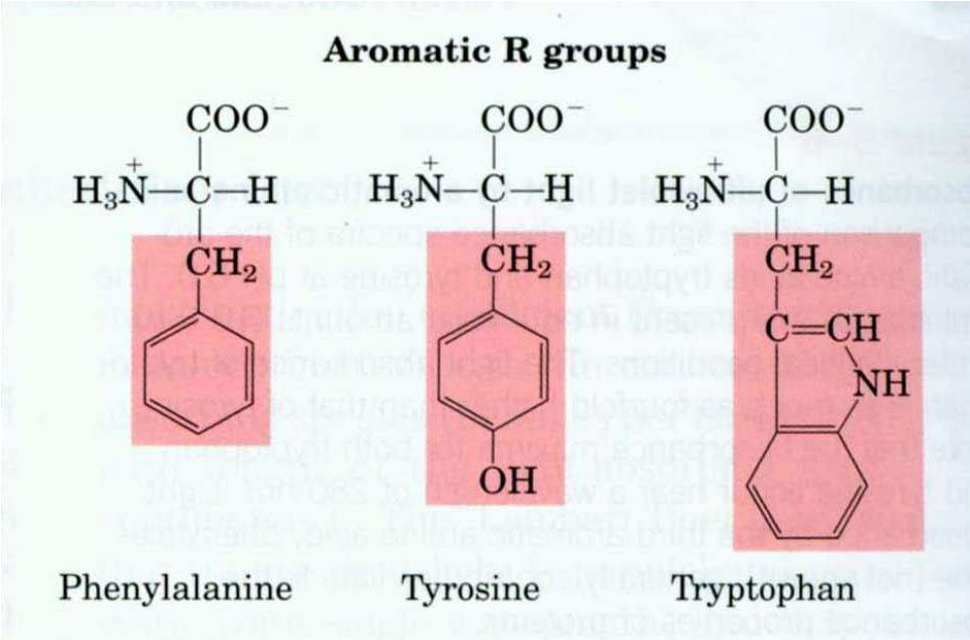
Disulfide bonds

- **-SH group of two Cys in proteins can be *oxidized to form a covalent disulfide bond.***
- **Disulfide bonds: play a special role in the structures of many proteins by forming covalent links.**



Aromatic amino acids

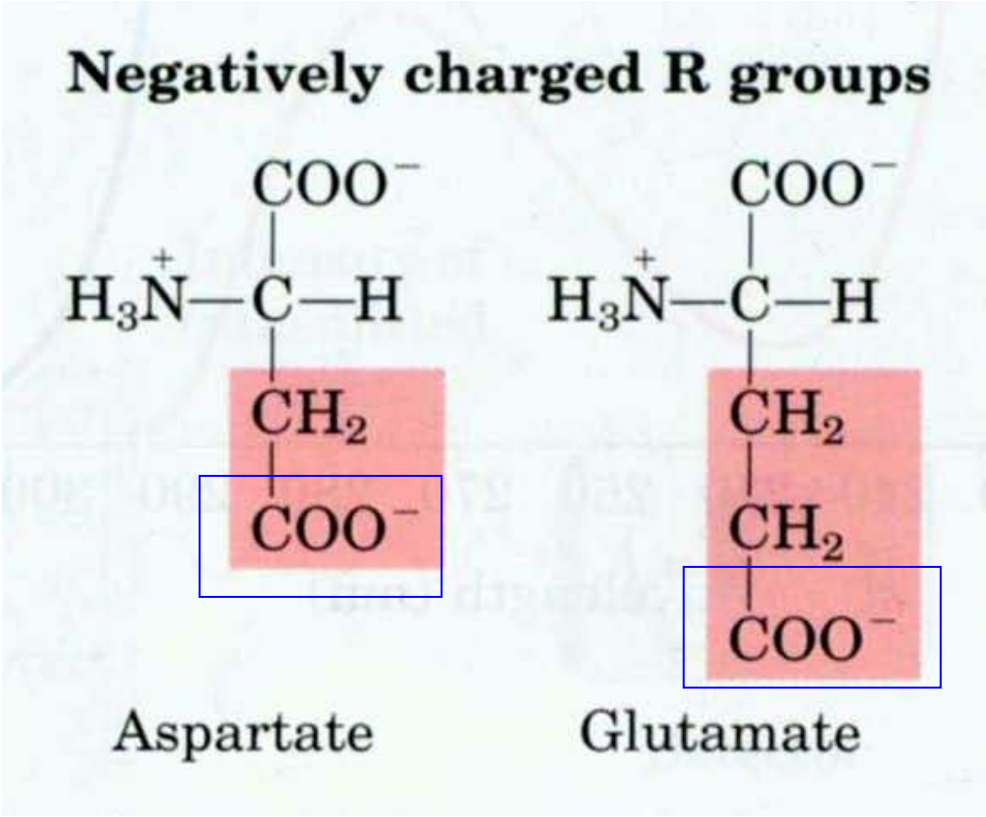
- **Phe, Tyr and Trp.**
  - **Phe and Tyr: benzene rings.**
  - **Tryptophan: indole ring.**



- **The -OH group in Tyr is an important functional group in proteins. (phosphorylation, hydrogen bond, etc), polar**

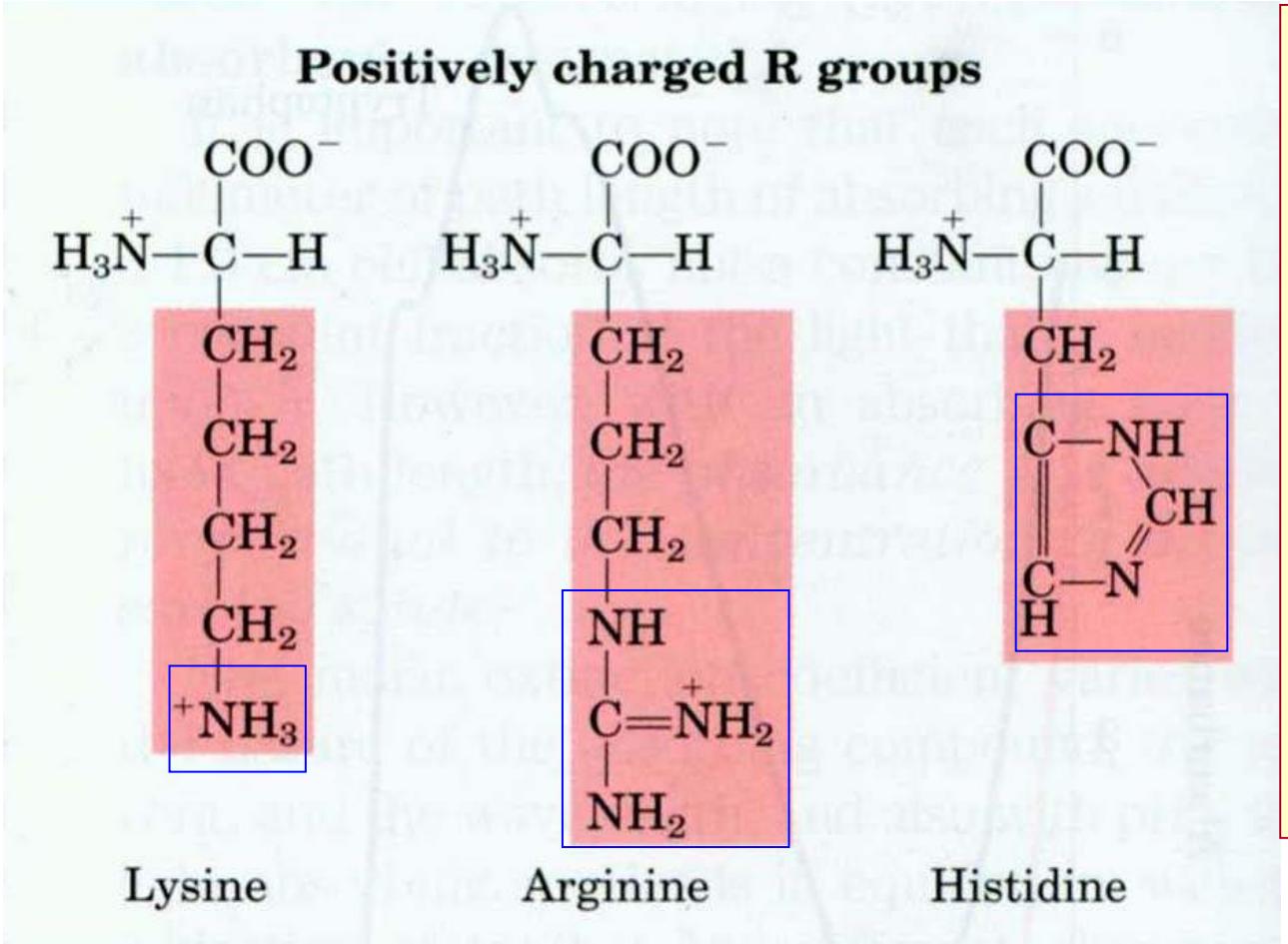
Acidic amino acids

- Asp and Glu
  - Have carboxyl in their R groups.



Basic amino acids

- Lys, Arg, and His.



- R groups
  - Amino
  - Guanidino
  - Imidazole
- Positive charged R groups at pH 7.0

- **Note these structural features**

1. All 20 are α-amino acids
2. For 19 of the 20, the α -amino group is primary; for proline, it is secondary (imino acid)
3. Except glycine, the α -carbons for 19 of them are asymmetric (or chiral).

- **Nonpolar Amino Acids**

Ala Val Leu Ile Gly Pro Phe Trp

- **Polar,Uncharged Amino Acids**

Ser Thr Cys Asn Gln Met Tyr

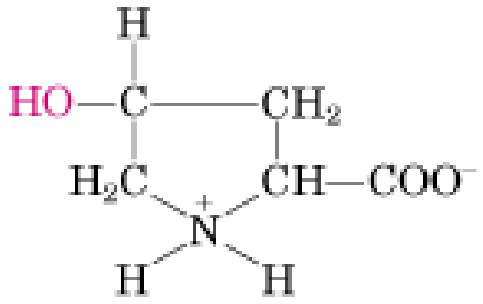
Aromatic

- **Acidic Amino Acids** Asp Glu

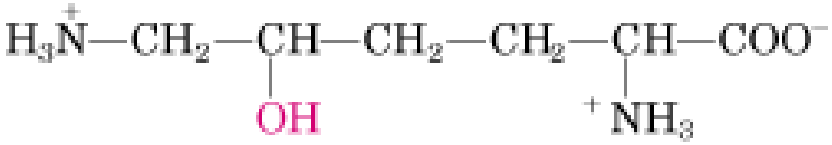
- **Basic Amino Acids** Lys Arg His

Nonstandard amino acids

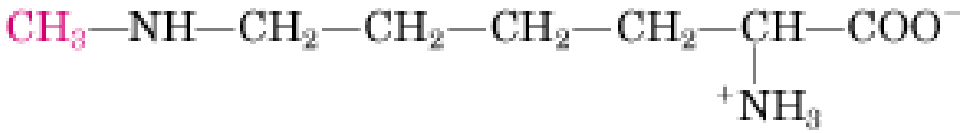
- **Amino acid derivatives found in proteins**
  - 4-Hydroxyproline **and 5-hydroxylysine in collagen.**
  - **6-*N*-Methyllysine in myosin.**



4-Hydroxyproline

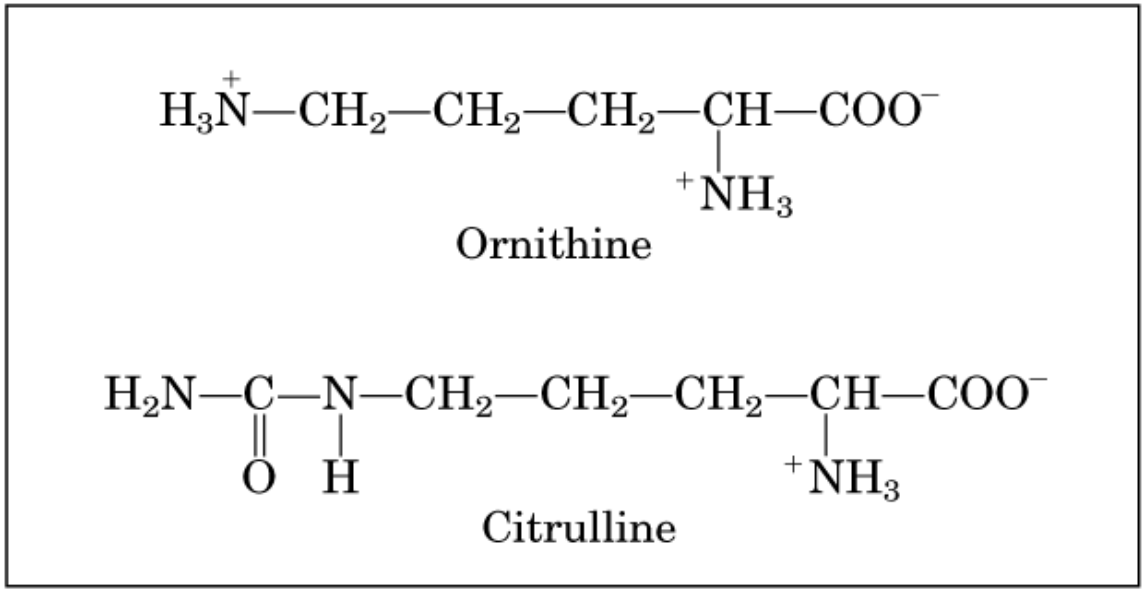


5-Hydroxylysine



6-*N*-Methyllysine

Many additional nonstandard amino acids are found in cells, but not in proteins



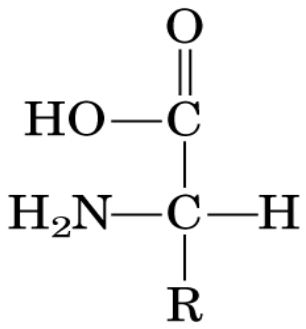
- Ornithine and citrulline
- Intermediates in amino acid metabolism.

Essential and non-essential amino acids

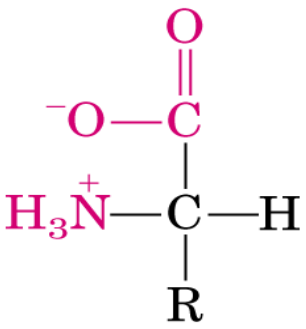
- Essential amino acids (or indispensable amino acids):
  - Cannot be synthesized by the humans, must be supplied in the diet
  - 8: Phe, Val, Thr, Trp, Ile, Met, Leu, Lys
- Semi-essential amino acids:
  - 2: His and Arg
  - Required by infants and growing children

Acid / Base properties of AAs

- Amino acid has both a basic amine group and an acidic carboxylic acid group.
- In neutral solution (pH 7.0), the amino acid contains a negative charge and a positive charge. It is called a *zwitterion* (German for “hybrid ion”).

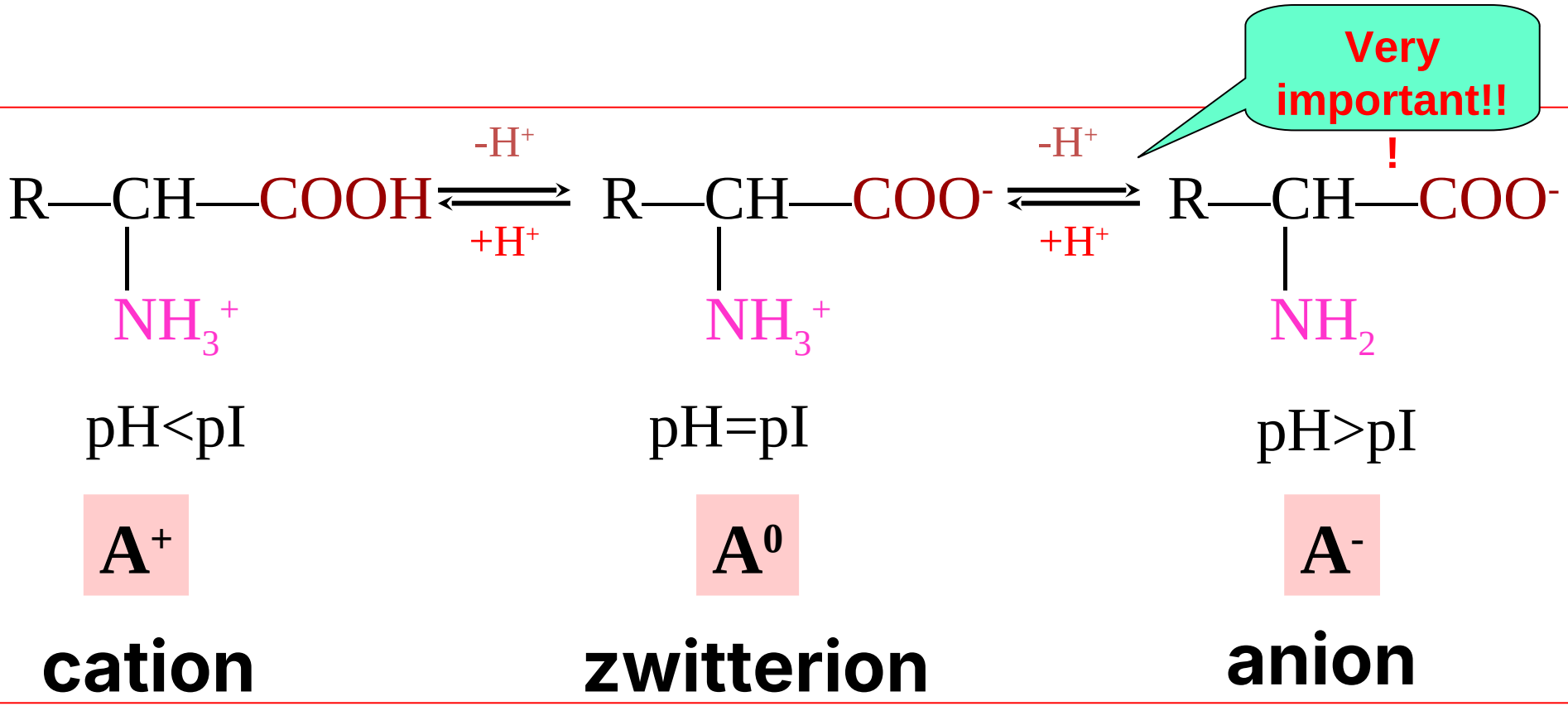


Nonionic form



Zwitterionic form

- AAs ionize to various states depending on pH values.
- pI: there is a specific pH (designated isoelectric point, pI) at which an AA has equal positive and negative charge (no net electric charge) .

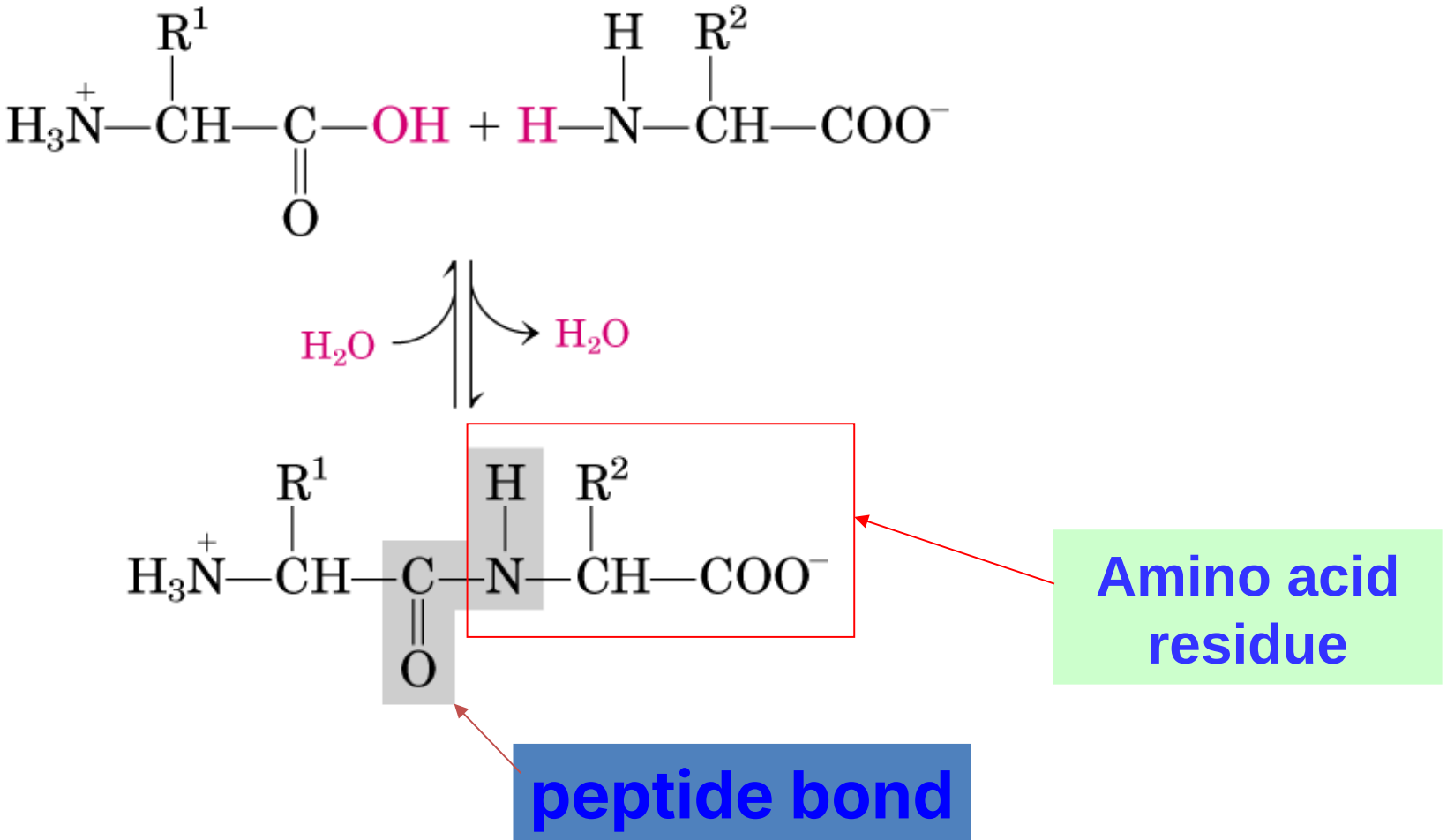


Properties and Conventions Associated with the Standard Amino Acids

Amino acid	Abbreviated names		M_r	pK_a values: pK_1 (—COOH)	pK_2 ($—NH_3^+$)	pK_R (R group)	pI	Hydropathy index*	Occurrence in proteins (%)†
Nonpolar, aliphatic R groups									
Glycine	Gly	G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala	A	89	2.34	9.69		6.01	1.8	7.8
Valine	Val	V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu	L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile	I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met	M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups									
Phenylalanine	Phe	F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr	Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp	W	204	2.38	9.39		5.89	−0.9	1.4
Polar, uncharged R groups									
Serine	Ser	S	105	2.21	9.15		5.68	−0.8	6.8
Proline	Pro	P	115	1.99	10.96		6.48	1.6	5.2
Threonine	Thr	T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine	Cys	C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn	N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln	Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged R groups									
Lysine	Lys	K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His	H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg	R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged R groups									
Aspartate	Asp	D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu	E	147	2.19	9.67	4.25	3.22	−3.5	6.3

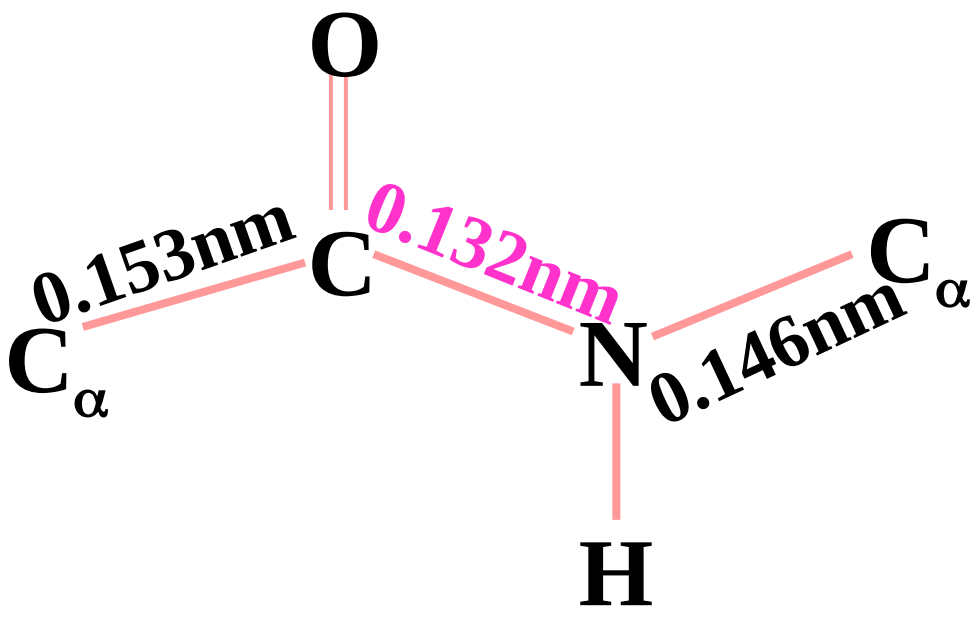
Polypeptides

Peptide bond: the special name given to the amide bond between the carboxyl group of one amino acid and the -amino group of another.

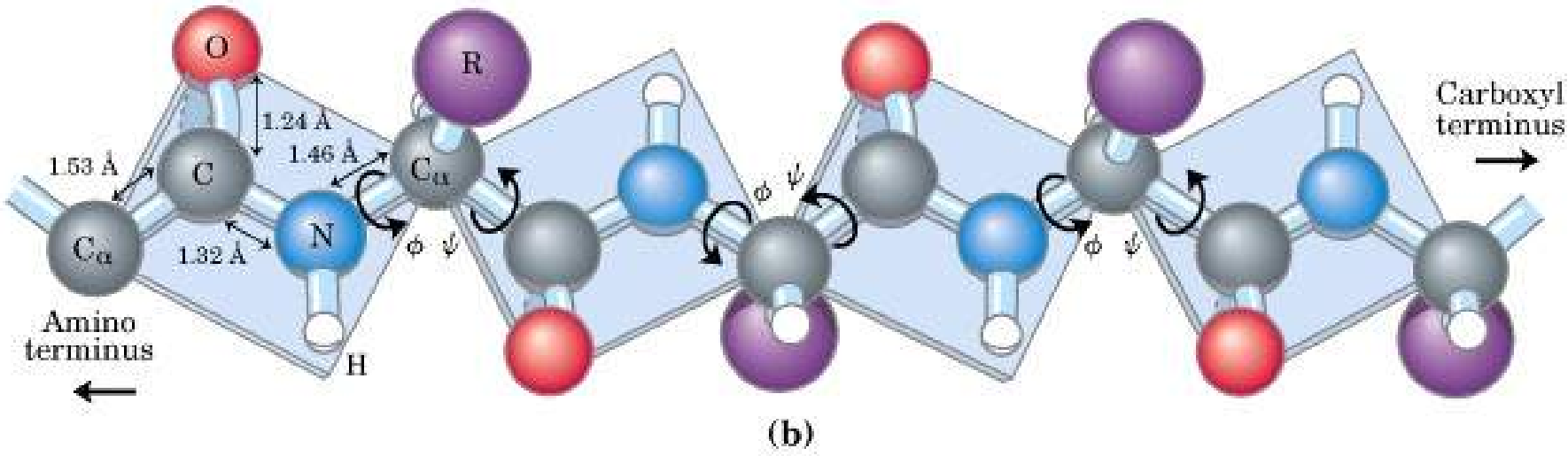


Characteristics of the peptide bond

- The peptide bond have partial double bond feature
  - about 0.132nm (C-N single bond, 0.149nm; C=N double bond, 0.127nm),
  - *rigid and unable to rotate freely*.

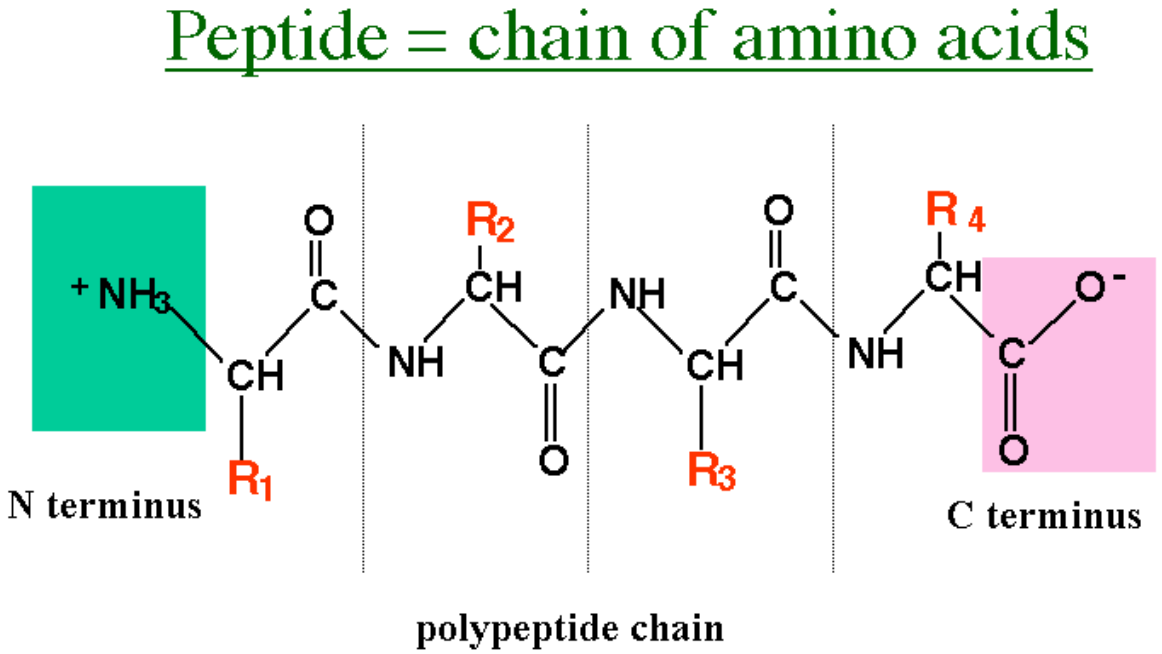


- The peptide bond is planar, trans-configuration and uncharged.
- Peptide plane: the six atoms attached to the peptide bond are coplanar.
- the carbonyl oxygen and the amide hydrogen are in trans positions.



The peptide chain is directional.

- Amino-terminal or N-terminal: the end having a free a-amino group.
- Carboxyl-terminal or C-terminal: the end having a free a-carboxyl group.
- By convention, the N-terminal is taken as the beginning of the peptide chain, and put at the left (C-terminal at the right).

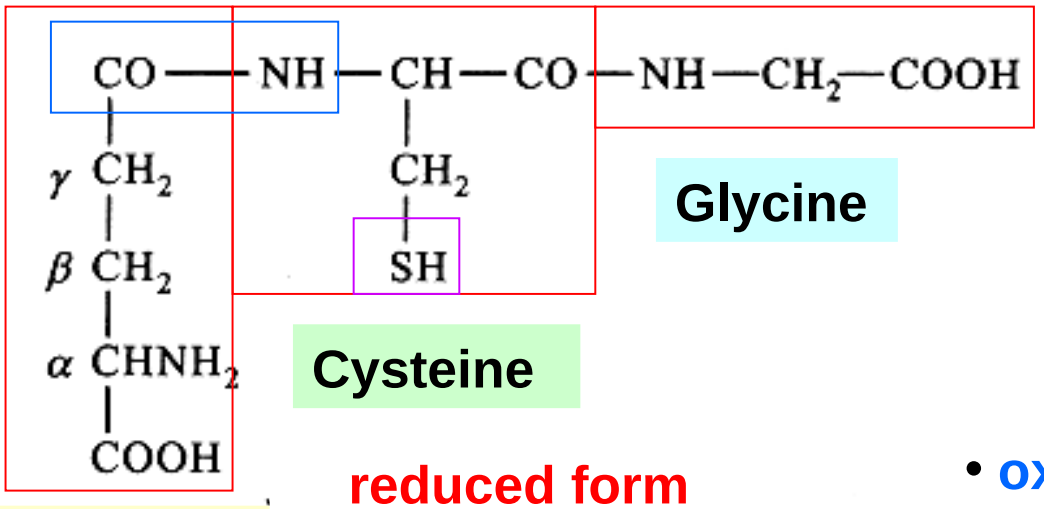


Polymers of amino acids

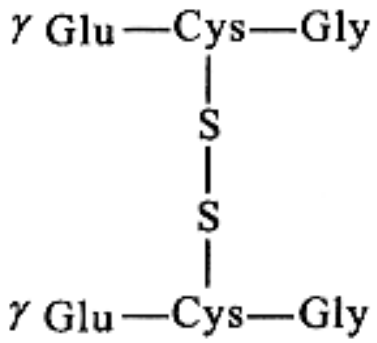
- Peptides can be classified according to how many amino acids they contain
  - Dipeptide: 2 amino acid residues, tripeptide: 3 residues, and so on
  - Oligopeptide: 12~20 residues
  - Polypeptide: many residues

Biologically important peptide

- Glutathione (GSH)
  - Tripeptide: glutamic acid, cysteine and glycine;
  - Function: important in biological oxidation-reduction reactions, has reduced and oxidized form.
  - It's the most important molecule you need to stay healthy and prevent disease.



reduced form
GSH



- oxidized form glutathione disulfide (GSSG)

CLASSIFICATION OF AMINO ACIDS BASED ON THEIR NUTRITIONAL VALUE

1-Essential amino acids ;They are those amino acids that cannot be synthesized in the body thus they are essential in the diet.

Essential amino acids include the following amino acids ;

- **Arginine***
- **Histidine***
- **Isoleucine**
- **Leucine**
- **Valine**
- **Lysine**
- **Methionine**
- **Threonine**
- **Phenylalanine**
- **Tryptophan**

2- Non-essential amino acids; They are those amino acids that can be synthesized in the body thus they are non-essential in the diet.

- **Alanine**
- **Asparagine**
- **Aspartate**
- **Glutamate**
- **Glutamine**
- **Glycine**
- **Proline**
- **Serine**
- **Cysteine (from Met*)**
- **Tyrosine (from Phe*)**

CLASSIFICATION BASED ON THEIR METABOLIC FATE

1-Glucogenic amino acids ;They are amino acids that can be used for glucose synthesis .

- **Aspartate**
- **Asparagine**
- **Arginine**
- **Phenylalanine**
- **Tyrosine**
- **Isoleucine**
- **Methionine**
- **Valine**
- **Glutamine**
- **Glutamate**
- **Proline**
- **Histidine**
- **Alanine**
- **Serine**
- **Cysteine**
- **Glycine**
- **Threonine**
- **Tryptophan**

CLASSIFICATION BASED ON THEIR METABOLIC FATE

2-Ketogenic amino acids; are those amino acids that can be used for ketone body synthesis only.

Lysine and Leucine.

3-Ketogenic and glucogenic amino acids ; They are those amino acids that can be used both for ketone body synthesis, and glucose synthesis.

- **Tryptophan**
- **Phenylalanine**
- **Tyrosine**
- **Isoleucine**
- **Threonine**

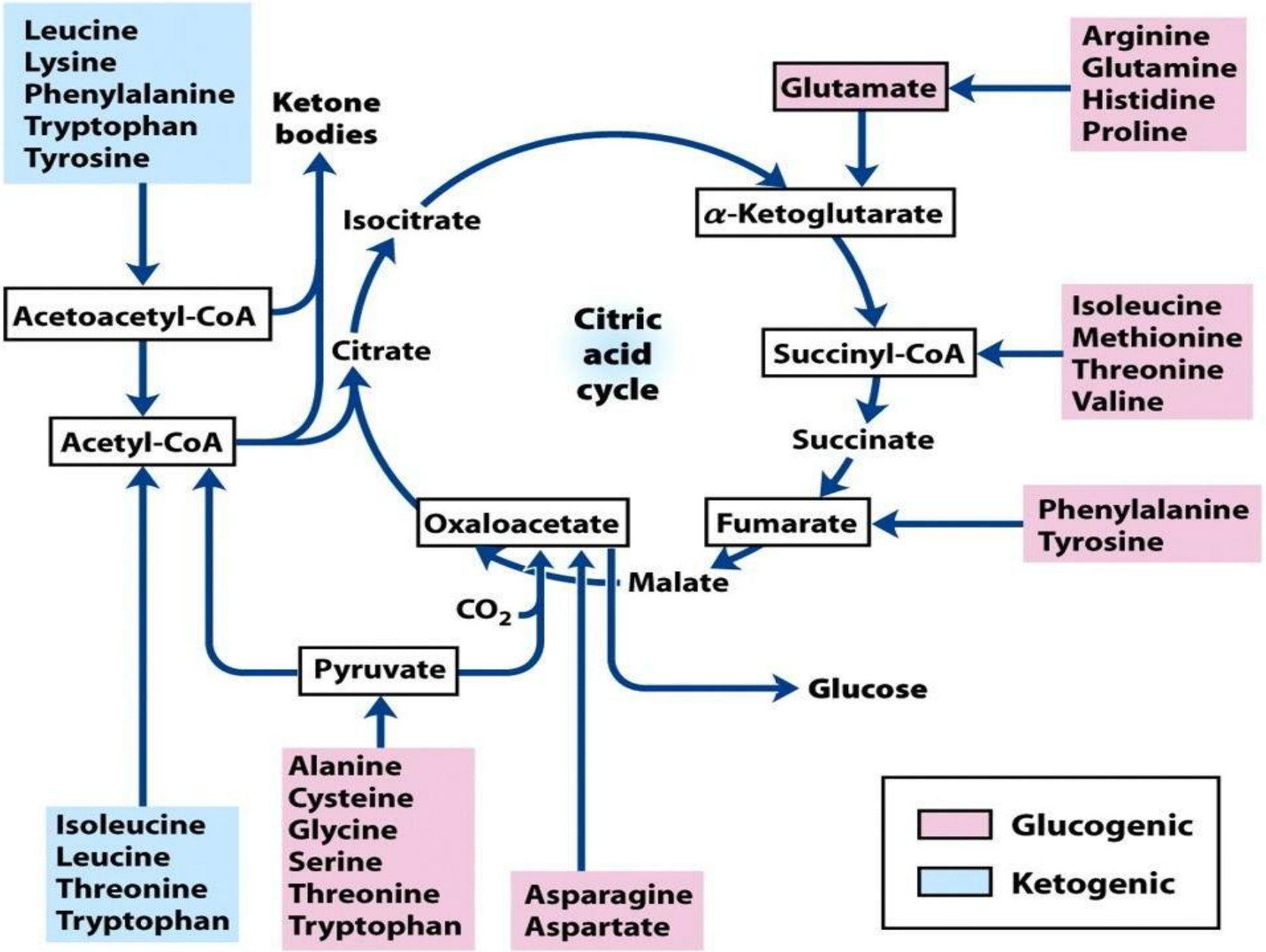


Figure 18-15
Lehninger Principles of Biochemistry, Fifth Edition
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Thank you